

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which expression is equivalent to $\frac{y+12}{x-8} + \frac{y(x-8)}{x^2y-8xy}$?

Answer

- A. $\frac{xy+y+4}{x^3y-16x^2y+64xy}$
- B. $\frac{xy+9y+12}{x^2y-8xy+x-8}$
- C. $\frac{xy^2+13xy-8y}{x^2y-8xy}$
- D. $\frac{xy^2+13xy-8y}{x^3y-16x^2y+64xy}$

Correct Answer: C

Rationale

Choice C is correct. Factoring the denominator in the second term of the given expression gives $\frac{y+12}{x-8} + \frac{y(x-8)}{xy(x-8)}$. This expression can be rewritten with common denominators by multiplying the first term by $\frac{xy}{xy}$, giving $\frac{xy(y+12)}{xy(x-8)} + \frac{y(x-8)}{xy(x-8)}$. Adding these two terms yields $\frac{xy(y+12)+y(x-8)}{xy(x-8)}$. Using the distributive property to rewrite this expression gives $\frac{xy^2+12xy+xy-8y}{x^2y-8xy}$. Combining the like terms in the numerator of this expression gives $\frac{xy^2+13xy-8y}{x^2y-8xy}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.